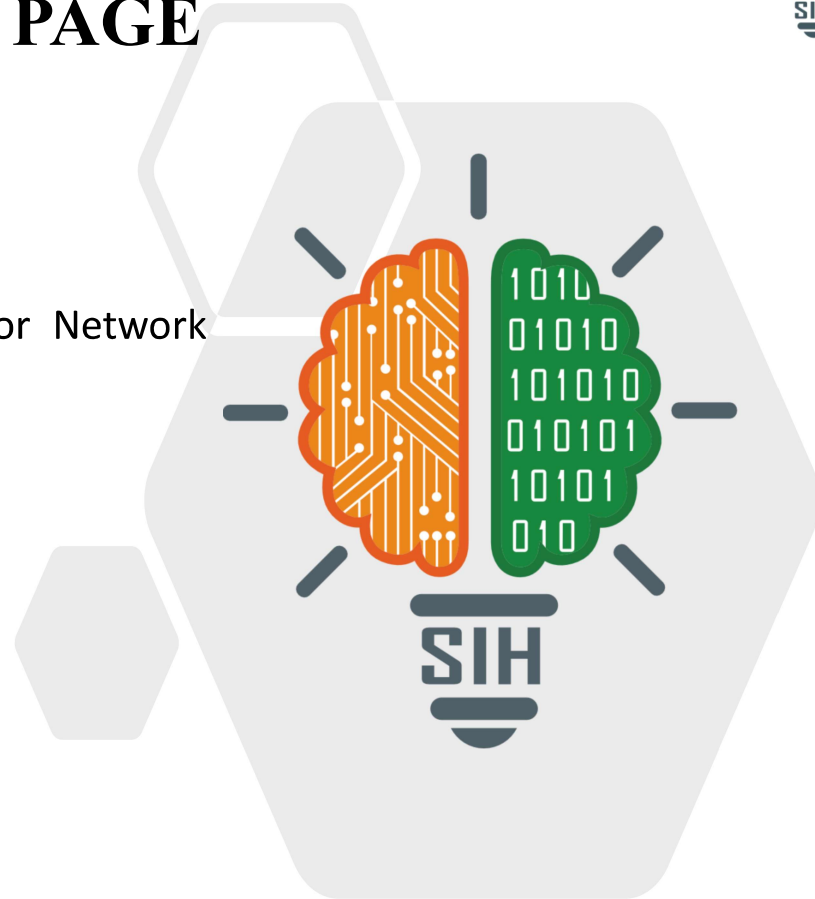


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TITLE PAGE



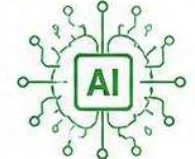
- **Problem Statement ID** –SIH26155
- **Problem Statement Title-** AI-Driven Multi-Vendor Network
Security Compliance Auditor
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name** – Cyber Shield

❖ **Proposed Solution (Describe your Idea/Solution/Prototype)****CyberConfig AI***AI-Powered Multi-Vendor Network Security Compliance Auditor*

"A vendor-agnostic AI auditor that **understands** network configurations, detects security/compliance gaps, **explains** risks, **recommends** fixes, **verifies** remediation, and **learns** from previously unseen configurations."

SOLUTION WORKFLOW**1. MULTI-VENDOR CONFIG INPUT**

Device configs, firewall rules, ACLs, protocols, policies, etc.

2. AI CONFIG UNDERSTANDING & NORMALIZATION

LLM/NLP parses and normalizes different vendor syntaxes into a common security model

3. SECURITY & COMPLIANCE ENGINE

Checks configuration against multiple frameworks and best practices

4. AI RISK ASSESSMENT

CRITICAL
HIGH
MEDIUM
LOW

Prioritized risks, severity scoring & compliance score calculation

5. EXPLAINABLE REMEDIATION

What is wrong
Why it is risky
Impact
Exact remediation commands

6. RE-SCAN & VERIFICATION

Apply fixes →
Re-scan → Verify
compliance
improvement

DETECT → **EXPLAIN** → **FIX** → **VERIFY** → **LEARN**

PROTOTYPE OUTPUT (DASHBOARD)

Devices Scanned	Compliance Score	Critical Issues	High Risk	Medium Risk
247	78%	12	31	47

TOP RISKS IDENTIFIED

Telnet Enabled	Risk: HIGH
Weak SSH Configuration	Risk: HIGH
Logging Disabled	Risk: MEDIUM
Over-Permissive ACL	Risk: MEDIUM

AI RECOMMENDATION (EXAMPLE)

Issue: Telnet Enabled on management interface
Impact: Unencrypted remote access, vulnerable to attacks
Fix: Disable Telnet service and enforce SSHv2

```
no telnet server enable
ip ssh version 2
```

UNIQUE INNOVATIONS

Self-Learning Parser
Learns unknown vendor syntax through expert feedback and improves over time.



Vendor-Agnostic Intelligence
Converts heterogeneous configs into a unified security model for accurate analysis.



Closed-Loop Security
Detect → Recommend
Fix → Re-scan → Verify
continuous compliance improvement.

REAL-WORLD IMPACT

Banks



Data Centers



Enterprises



Government



Hospitals

- ✓ Reduce manual auditing time
- ✓ Improve compliance posture
- ✓ Prioritize critical risks
- ✓ Accurate, explainable & actionable insights
- ✓ Continuous learning & future-ready

◆ TECHNICAL APPROACH — CYBERCONFIG AI

Vendor-agnostic • Explainable AI • Policy-as-Code • Closed-loop remediation

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1 INGEST & NORMALIZE

CONFIGURATION FABRIC

- Cisco • Juniper • Fortinet • Palo Alto
- CLI / API / backup configs
- Unknown vendor discovery
- Canonical security model

2 UNDERSTAND

AI NORMALIZATION

- Parser + LLM/NLP pipeline
- Syntax → semantic intent
- Vendor dialect normalization
- Context-aware feature extraction

3 COMPLY & SCORE

COMPLIANCE INTELLIGENCE

- CIS • NIST • STIG policies
- Policy-as-Code engine
- Control mapping & evidence
- Weighted compliance scoring

4 EXPLAIN & REMEDIATE

RISK & EXPLAINABILITY

- Risk = severity × exposure × impact
- Criticality prioritization
- WHY / WHAT / IMPACT
- Confidence + evidence trail

5 VERIFY & LEARN

SAFE REMEDIATION

- Generate exact fix commands
- Dry-run / approval gate
- Apply → re-scan → compare
- Rollback on regression

6 GOVERN

GOVERNANCE

- RBAC
- Audit log
- Versioning
- Human approval

AI DECISION LAYER

Hybrid Intelligence

LLM/NLP interprets unseen syntax and context;
deterministic rules enforce security controls.
→ AI proposes • Policy engine validates

RISK ENGINE

Evidence-Based Prioritization

Asset criticality + exploitability + exposure
+ control severity → explainable risk score
→ Critical / High / Medium / Low

REMEDiation GUARDRAILS

Human-in-the-Loop Security

Recommendation → dry-run → approval → apply
→ verification → rollback if regression
No blind autonomous changes.

CONTINUOUS LEARNING

Closed-Loop Feedback

Analyst feedback + remediation outcome
→ parser/policy improvement
→ better future detection

IMPLEMENTATION & PROTOTYPE

Prototype stack

Frontend: React + TypeScript
Backend: FastAPI + Python
AI: LLM/NLP + embeddings • Rules: Python/OPA-style policies
Storage: PostgreSQL + vector index • Deployment: Docker

UNIQUE VALUE

What makes it different

1. Self-learning vendor parser for unseen syntax
2. Policy-as-Code + LLM = controlled AI
3. Closed-loop Detect → Explain → Fix → Verify → Learn
4. Evidence-backed recommendations, not black-box scores

DEMO SUCCESS METRICS

Compliance
Before → After

Risk
Prioritized

Fix
Exact command

Proof
Re-scan evidence

FEASIBILITY & VIABILITY

Practical, Secure, Scalable and Ready for Real-World Deployment



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SOLUTION FEASIBILITY – AT A GLANCE



Uses existing network data (read-only)



Proven AI/ML + NLP technologies (LLM + RAG)



Policy-as-Code compliance (CIS/NIST/STIG)



Dry-run, approval & rollback for safe remediation



Lightweight, modular & scalable design



Pilot-ready MVP with measurable impact

TECHNICAL FEASIBILITY



- Read-only access via SSH/API/Export (no production change).
- Multi-vendor parsing using NLP + templates.
- AI-driven risk analysis using policy engine.
- Prototype built with open-source stack.

OPERATIONAL VIABILITY



- Designed for SOC, Network Operations & Audit teams.
- Simple UI, actionable insights, less manual effort.
- Integrates with existing tools (SIEM, ITSM, CMDB).
- Continuous learning reduces future effort.

SECURITY & TRUST



- Read-only ingestion.
- All data encrypted in transit & at rest.
- Role-based access control (RBAC).
- Full audit trail, versioning & change logs.

SCALABILITY & COST



- Horizontal scaling using microservices.
- Cloud or on-prem deployment.
- Cost-effective using open-source stack.
- High ROI through risk reduction & automation.

IMPLEMENTATION ROADMAP

01



LAB BUILD
(0–4 Weeks)

Core modules, vendor parsers, policy engine, UI prototype.

02



PILOT DEPLOYMENT
(1–2 Months)

Selected devices & locations, collect feedback & tune AI.

03



CONTROLLED BOLLOUT
(3–6 Months)

Organization-wide deployment with monitoring & training.

04



SCALE & OPTIMIZE
(6–12 Months)

Advanced analytics, self-learning & cross-framework coverage.

05



NATIONAL SCALE
(12+ Months)

Multi-tenant, highly scalable, continuous innovation.

END-TO-END WORKFLOW (CLOSED LOOP)



1. INGEST

Configs from multi-vendor devices



2. ANALYZE

AI understands configs & maps to policies



3. DETECT & EXPLAIN

Misconfigurations with risk, impact & rationale



4. RECOMMEND

Actionable fixes with severity priority



5. APPROVE & REMEDIATE

Dry-run → Approve → Apply (Safe Remediation)



6. RE-SCAN & VERIFY

Verify compliance & generate evidence



7. LEARN & IMPROVE

Feedback loop improves future detections

RISK & CHALLENGES → MITIGATION

POTENTIAL CHALLENGE	MITIGATION STRATEGY	RESIDUAL RISK
</> Unseen vendor syntax	Self-learning parser + human validation	Low
AI hallucinates / wrong suggestion	RAG + policy engine + human-in-loop approval	Very Low
Configuration change risk	Dry-run, rollback, backup & staged rollout	Very Low
Data sensitivity	Encryption, RBAC, read-only access	Very Low
Scalability with large networks	Microservices, caching, async processing	Low

PROTOTYPE VALIDATION & SUCCESS METRICS



ACCURACY

≥ 90% detection accuracy on benchmark configs



TIME SAVED

≥ 70% reduction in audit & remediation time



COVERAGE

Multi-vendor + CIS/NIST/STIG compliance



RISK REDUCTION

High-risk misconfigurations reduced by ≥ 80%



AUDIT READINESS

Evidence report generated automatically



USER ADOPTION

High usability & actionable insights

IMPACT, UNIQUENESS & DEPLOYMENT

From **configuration audit** → **actionable security** → **continuous compliance**

1. KEY DIFFERENTIATORS

-  **Vendor-Agnostic:** Cisco, Juniper, Fortinet, Palo Alto & unknown vendors
-  **Explainable AI:** Every finding includes Risk → Reason → Impact → Fix
-  **Closed-Loop Verification:** Detect → Recommend → Approve → Remediate → Re-scan
-  **Self-Learning Parser:** Adapts to previously unseen configuration syntax
-  **Policy-Aware AI:** CIS + NIST + STIG rules combined with organizational policies

2. REAL-WORLD IMPACT


70%+

Audit & manual
review time
reduction


80%+

High-risk
misconfiguration
reduction


90%+

Target detection
accuracy


100%

Evidence-backed
audit trail

3. WHO BENEFITS?

- | | | | | |
|---|---|---|--|--|
| 
Defence & Government
STIG / NIST compliance | 
Banks & BFSI
Reduce attack surface | 
Enterprises
Multi-vendor security visibility | 
Data Centers
Continuous configuration assurance | 
SOC Teams
Faster triage & remediation |
|---|---|---|--|--|

4. DEPLOYMENT MODEL

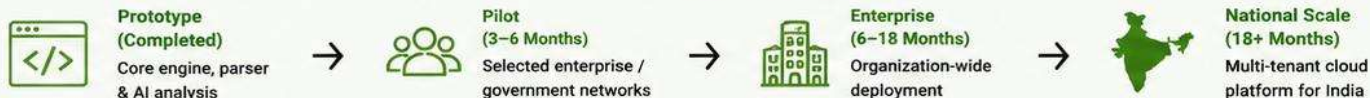


5. WHY CYBERCONFIG AI?



Unlike conventional configuration scanners, **CyberShield AI** does not only detect misconfigurations – it explains their security impact, recommends policy-aware remediation, verifies the fix, and learns from new vendor syntax.

PATH TO SCALE



LONG-TERM VISION

AI-powered national platform for secure, compliant & resilient network infrastructure.



Research & References:

1. Secure Military Communication Platforms

• **SAMBHAV Platform** – Indian Army's indigenous secure communication system.

<https://www.gktoday.in/sambhav-secures-military-communications/>

• **Signal Protocol** – Open-source E2E encryption standard used globally.

https://en.wikipedia.org/wiki/Signal_Protocol

• **Wickr RAM** – DOD-approved enterprise messaging for military use.

<https://www.chanty.com/blog/dod-approved-messaging-apps/>

2. Encryption & VPN Integration

• **Signal Security Architecture** – Double Ratchet Algorithm and key agreement protocols.

<https://profincoito.me/blog/security/signal-security-architecture/>

• **Military Communication Security** – Best practices for defence messaging platforms.

<https://saltcommunications.com/news/beyond-the-frontlines-safeguarding-military-operations-through-secure-communication/>

3. Zero-Knowledge & Privacy Standards

• **End-to-End Encryption Standards** – Implementation of E2E with administrative controls.

<https://www.rocket.chat/blog/military-messaging>

• **VPN Tunneling Protocols** – IPsec and OpenVPN for military-grade network security

4. Compliance & Audit Framework

• **Military Communication Compliance** – Audit trails and forensic capabilities for defence systems

• **Data Sovereignty Standards** – India-hosted infrastructure requirements for national security

Market Benching:

1. SAMBHAV (Indian Army) What it does: Indigenous military communication with multi-layer encryption for 30,000+ Army devices. **Why our platform:**

- Extends to veterans and defence families (AMBHAV is Army-only)

- Comprehensive audit trails and compliance reporting

2. Signal What it does: Open-source E2E encrypted messaging serving 100M+ global users. **Why our platform:**

- HQ-controlled membership vs phone number registration

- Administrative oversight (Signal has no admin controls)

- VPN-layer protection on top of E2E encryption

3. Wickr RAM What it does: DOD-approved enterprise messenger for US military/government. **Why our platform:**

- India-hosted sovereign infrastructure vs foreign servers

- Designed for Indian defence ecosystem and families